

SURVEY ARTICLE

Safety in Robotic Haircutting

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ABSTRACT

Robotic haircutting is an emerging application of service robotics in which safety is paramount. Unlike conventional collaborative or domestic tasks, haircutting requires prolonged operation near the human head, one of the most vulnerable body regions, and often involves sharp or heated tools. These characteristics make safety central to both functional reliability and user acceptance. This paper reviews the safety landscape of haircutting robots by examining risks across the full task workflow and synthesizing lessons from related domains, including collaborative robotics, healthcare robotics, and autonomous systems. It proposes a three-layer framework to organize existing approaches and illustrate how safety can be embedded from hardware design to control logic. In addition, a series of proof-of-concept demonstrations illustrates the practical implementation of the proposed three-layer safety framework in representative haircutting scenarios. The paper further highlights opportunities and unresolved challenges in ensuring safe human-robot interaction in haircutting contexts, providing guidance for future research and deployment.

1 | Introduction

Human-robot interaction (HRI) has become a central research topic as robots increasingly operate in collaborative, healthcare, and domestic environments. Ensuring safety in such interactions is critical, since physical contact may involve sensitive body regions where even minor design flaws can lead to discomfort or injury. Consequently, safety has received increasing attention in both engineering practice and regulatory development.

Existing studies on safe HRI have primarily focused on collaborative industrial robots, healthcare robots, and service robots. On the one hand, collaborative robots highlight intrinsic mechanical compliance, force limiting, and standardized safety modes to enable shared workspaces with human operators (Cherubini et al. 2016). On the other hand, healthcare and rehabilitation robots introduce compliant actuation, adaptive control, and fault detection to safeguard patients in close-contact scenarios (Mohebbi 2020). Moreover, service robots

address ergonomic design, human-aware navigation, and multimodal perception to support dependable operation in unstructured environments (Haddadin and Croft 2016). These research directions provide a valuable foundation but remain general in scope.

Recent studies have highlighted robotic haircutting as a distinctive class of service robotics in which safety requirements are particularly demanding (Li 2025a). Unlike most collaborative or domestic applications, haircutting tasks require continuous and precise manipulation around the head, one of the most sensitive body regions (Khan and Li 2025). It also involves sharp or heated tools, such as clippers, scissors, and dryers, which pose direct physical hazards (Li 2025b). Beyond physical safety, haircutting outcomes must satisfy esthetic requirements, introducing task-specific constraints rarely encountered in conventional HRI domains. Although existing reviews summarize system feasibility and design aspects, they lack insight into the risks inherent in haircutting, making comprehensive hazard assessment and mitigation difficult. Together, these

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characteristics make robotic haircutting a particularly challenging and underexplored application.

Despite progress in HRI research, systematic approaches to haircutting safety are still missing. To fill this gap, this paper provides the first systematic review of safety in robotic haircutting, highlighting the unique challenges of operating around sensitive head regions with sharp or heated tools. It presents a structured analysis of risks across the haircutting workflow, consolidating hazards from the perception, planning, execution, and feedback stages. Building on these insights, a three-layer perspective comprising the mechanical, sensing, and algorithmic layers is proposed to organize safety strategies and illustrate how solutions from related domains can be adapted to haircutting robots. This review further identifies open research gaps and outlines directions for future studies, experimental validation, and the development of domain-specific standards toward safe real-world deployment. Given the limited haircutting-specific experimental evidence currently available, this survey does not aim to establish application-validated safety limits for robotic haircutting. Instead, it synthesizes safety knowledge from adjacent domains and provides preliminary design references and future validation directions.

The remainder of this paper is organized as follows. Section 2 reviews background and related work on HRI safety, highlighting prior studies in perception, hardware compliance, and human factors. Section 3 analyzes the workflow of haircutting robots and categorizes safety risks across stages and system components. Sections 4 to 6 provide an in-depth discussion of the proposed three-layer safety framework: the mechanical layer, the sensing layer, and the algorithmic layer. Section 7 presents a series of proof-of-concept demonstrations to illustrate how the proposed three-layer safety framework can be applied in a haircutting scenario. Section 8 identifies current research gaps and outlines directions for future work. Finally, Section 9 concludes this paper.

2 | Background and Related Work

Safety in HRI has been investigated from multiple complementary perspectives, as robots are increasingly expected to operate in human-centered and unstructured environments. At an early stage, an atlas of HRI was presented (De Santis et al. 2008), identifying mechanical design, actuation, and control architectures as critical determinants of safety and dependability, while highlighting the need for evaluation metrics beyond cognitive-level considerations. In parallel, (Alami et al. 2006) surveyed the state of the art in safe and dependable HRI, emphasizing an integrated view that combines mechanical compliance, intrinsic safety design, fault tolerance, and sensing reliability within the PHRIDOM and PHRIENDS projects. More recently, a study (Selvaggio et al. 2021) examined the role of autonomy in HRI, differentiating shared control from shared autonomy paradigms and highlighting the risks involved in arbitrating human and robot authority, especially in dynamic or uncertain environments. In addition, some studies have analyzed injury mechanisms and severity indices by borrowing criteria from automotive crash testing, such as the Head Injury Criterion (HIC), and extending them to robotic contacts with different body parts (Kirschner et al. 2024; SMBPB et al. 2025).

Across these studies, common concerns can be identified: mechanical properties often determine the severity of physical contact, control and autonomy strategies are essential for regulating interaction forces, and system dependability relies on sensing reliability and fault tolerance.

2.1 | Haircutting-Related System Overview

We first examine existing efforts in haircutting and hair-related robotic systems, which have primarily appeared in patents, media demonstrations, and academic research prototypes. To ensure broad coverage, the survey was conducted across multiple sources, including patent databases (e.g., Google Patents), academic platforms such as Web of Science, Elsevier, and Google Scholar, as well as publicly available media and social platforms (e.g., YouTube) where early-stage robotic haircutting systems are often showcased. The search keywords included “haircutting robot”, “barber robot”, “robotic haircutting,” and “automatic haircut,” as well as related variants.

Despite this multi-channel search strategy, the number of relevant works remains remarkably limited. Robotic haircutting concepts can be traced back to the early 1980s (Herman 1986), yet only a small set of representative patents, demonstrations, and research prototypes have been reported since then. This scarcity reflects the technical and safety challenges of the domain. Accordingly, rather than aiming for exhaustive enumeration, representative examples (summarized in Table 1) are selected to highlight a consistent gap: existing systems predominantly emphasize functionality or proof-of-feasibility, while systematic and quantitative safety considerations remain largely absent.

Patents mostly describe self-service or semi-automatic devices that shield cutting tools with chambers or simple mechanical guards. These designs focus on basic functionality and convenience, while safety remains limited to passive protection without sensing or pHRI-related control. Media demonstrations typically present robotic haircutting as entertainment or publicity, such as automated shaving or pop-up robotic barbers. They emphasize novelty rather than technical rigor, and safety is mentioned only in broad terms. Research prototypes use robotic arms or vision-guided systems, but few address haircutting directly. Most target hair-care tasks such as combing or strand detection. Although these systems explore head-area interaction, they rarely include force limits, tool-skin distance control, or risk analysis for sharp tools.

Overall, existing systems demonstrate feasibility but lack comprehensive, quantitative safety analysis. This gap underscores the need for a dedicated safety framework tailored to robotic haircutting.

2.2 | Perception and Cognition Safety

In HRI, perception and cognition are primarily concerned with detecting proximity, contact, and human motion to prevent harmful collisions. Early approaches emphasized vision-based human detection (Halme et al. 2018) and skeletal tracking with RGB-D cameras (Secil and Ozkan 2022) to estimate body joint

TABLE 1 | Summary of existing haircutting and haircutting-related systems across patent, media, and research channels.

Channel	Purpose	Safety discussion	Ref
Patent	One of the earliest “robotic haircutting” patents; proposes a mechanical arm that moves around the user’s head to cut hair.	Relies only on physical enclosure; lacks sensing and control-level safety.	Herman (1986)
Patent	A self-service rotating-blade haircutting device designed for users to cut their own hair.	No safety mechanisms; the user directly interacts with high-speed cutting blades.	Köhler (2007)
Patent	A haircutting system that includes head localization and adjustable cutting-length control.	Uses only mechanical guards; does not discuss pHRI-related safety.	Weinman (2013)
Patent	A robotic-arm haircutting system using 3D head modeling and trajectory planning.	Mentions avoiding contact with the skin but provides no quantitative safety criteria.	Chen (2023)
Media	Demonstrate extreme automation/marketing stunt of head-shaving.	Reports note that the helmet completed shaving in 20 s without cuts or scratches on the volunteer’s scalp.	The Shaving Helmet (n.d)
Media	Public demo of robotics in a non-traditional barber.	Coverage mainly focuses on the charity event; no explicit mention of safety measures or risk control.	NBC News Technology Desk (n.d)
Media	Exploratory robotic haircut trimming for neck/upper hair.	No identifiable explicit discussion of safety in available sources.	Buzz Robotics (n.d)
Media	Proof-of-feasibility/entertainment of automated haircut.	Robot measures its distance from the scalp to avoid accidental cuts.	Stuff Made Here (n.d)
Research	Developed a camera-guided robotic system for self-haircutting using head tracking.	No contact-safety modeling; lacks force control and tool-skin distance constraints; no discussion of pHRI risks.	Kyosuke et al. (2014)
Research	Built an automated hair-combing robot using visual strand detection and path planning.	Focuses on smooth interaction and user comfort; pertains only to combing with no blades, thus lacking haircut-specific safety design.	Dennler et al. (2021)
Research	Implemented a robotic system that repositions front hair using root-based path planning.	Emphasizes control accuracy only; no force limits, no skin-distance modeling, and no risk controls for sharp tools or potential injury.	Kim et al. (2025)
Research	Developed a soft, sensor-integrated robot for safe, low-force hair-care interactions.	One of the few works explicitly addressing low-force scalp contact; still not a haircutting scenario and lacks safety modeling for cutting tools.	Yoo et al. (2025)

positions and ensure minimum safe distances between humans and robots. To address occlusions and sensor noise, recent work has developed multimodal sensing and fusion methods, integrating vision, depth, tactile, and force/torque sensors to provide redundancy and cross-validation (Xue et al. 2020; Papanastasiou et al. 2019). Such redundancy is critical in physical contact scenarios, where failures in detection may result in unsafe interaction forces.

Beyond raw perception, cognition in HRI often requires predicting human trajectories and estimating intent. Human modeling studies, particularly those combining musculoskeletal dynamics and sensorimotor control, have been employed to anticipate human motion and adjust robotic responses in real time (Fang et al. 2023). Similarly, multimodal frameworks fusing biosignals (e.g., electromyography, EMG and electroencephalography, EEG) with motion data have been proposed for intent recognition and adaptive assistance. These efforts illustrate a growing trend toward

combining low-level sensing with higher-level inference to achieve robust perception-cognition pipelines.

2.3 | Action and Hardware Safety

Once perception and cognition establish reliable information about human proximity and intent, the hardware and control layer serves as the final safeguard in HRI. Research has shown that compliance control methods, such as impedance and admittance control, are essential for modulating interaction forces and enabling safe execution during collisions (Ahmed 2011). In healthcare and rehabilitation robotics, collision detection and reflexive reactions have been widely investigated to ensure safe operation near sensitive body regions (Iroju et al. 2017).

In parallel with control approaches, mechanical compliance has been introduced through soft robotics, viscoelastic covers, and

series elastic actuators, which help absorb impact energy and reduce peak collision forces (Polygerinos et al. 2017; Qaddoori Fenjan and Fathollahi Dehkordi 2024; Mohammadi Nejad Rashty et al. 2024). Recent developments in semi-active compliant actuation, such as magneto-rheological fluid-based joints, have demonstrated the ability to combine adaptability with intrinsic safety, offering reliable protection during both static and dynamic collisions (Ahamed et al. 2018). These results support a dual strategy of compliant actuation and adaptive control as an effective paradigm for mitigating risks in HRI.

Nevertheless, even when compliant actuation and adaptive control are technically effective, the practical acceptance of HRI systems further depends on human factors and compliance with system-level regulations.

2.4 | Human and Systemic Factors

Beyond technical measures, HRI safety also depends on how humans perceive safety and how systems comply with regulatory and organizational frameworks. Human trust and comfort are shaped not only by objective limits on force and pressure but also by subjective factors such as predictability, controllability, and cultural acceptance (Law and Scheutz 2021; Hancock et al. 2011; Yagoda and Gillan 2012). Prior studies have shown that perceived safety may deviate substantially from physical safety, particularly when interaction involves sensitive body regions such as the head and neck (Rubagotti et al. 2022). This discrepancy underscores the importance of ergonomic design and transparent robot behavior in safety evaluations.

To capture these interrelated dimensions, Figure 1 illustrates a conceptual framework that integrates regulatory standards, human factors, and systemic issues. In this framework, regulatory standards provide objective references for safety limits, human factors determine public acceptance of robots, and systemic issues address deployment aspects such as certification, liability, and maintenance. Together, these three domains determine whether physical safety can be effectively translated into user trust and social acceptance.

Building upon this conceptual view, Table 2 summarizes the main international standards and directives that shape safety

requirements for haircutting robots. Standards such as ISO/TS 15066, ISO 13482, and ISO 10218 address collaborative operation, personal care robotics, and industrial robot safety, while ISO 18646 and IEC 80601 extend requirements to service and domestic environments. In addition, ISO 14971 and the EU Machinery Directive provide structured approaches to risk management, certification, and liability allocation. Together, these standards define both the technical criteria for robot safety and the procedural obligations for compliance in real-world deployment.

When these general standards are applied to robotic haircutting, however, their guidance becomes limited by the task-specific nature of close-contact hair manipulation. The current standards landscape provides only partial guidance for ensuring haircutting safety. ISO 13482 offers the foundation for personal-care robots that operate close to the user and therefore covers general requirements for physical interaction, protective design, and safe use in everyday environments. ISO/TS 15066 contributes principles for limiting forces and pressures during collaborative contact, which helps define safe operation when the robot works near sensitive areas such as the head and face. These standards establish an initial regulatory pathway for future commercial products, yet they leave important aspects of haircutting task unaddressed. Neither standard considers the behavior of sharp tools, the dynamics of trimming close to the scalp, or the possibility of hair entanglement. Because these elements are central to haircutting, any attempt to certify a commercial system under frameworks such as the EU Machinery Directive would require additional task-specific evidence, including measurements of scalp contact forces, validation of tool safety, and analyses of failure modes that are not covered by existing standards.

In summary, perception and cognition, action and hardware, as well as human and systemic considerations represent three interconnected perspectives of HRI safety. Prior studies in collaborative, service, and medical robots provide foundational principles, including multimodal sensing, compliant control, soft materials, and standardized force thresholds. However, haircutting robots simultaneously face unique challenges across all three domains: fine-grained perception of hair and skin, continuous close-contact manipulation with diverse tools, and

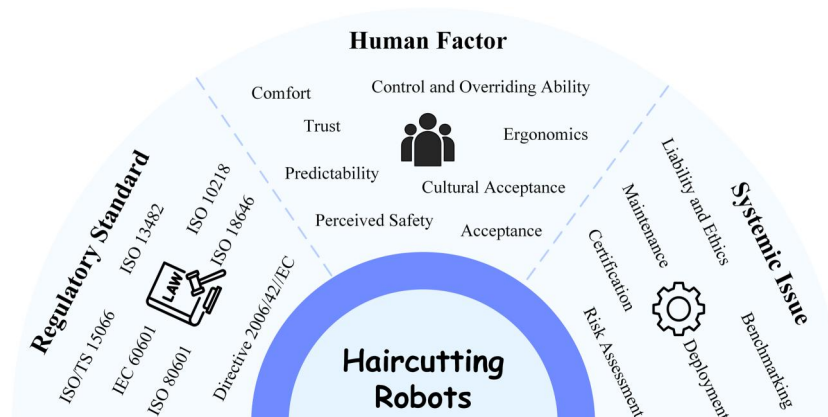


FIGURE 1 | Conceptual framework of human and systemic factors for haircutting robots. Safety depends on the integration of regulatory standards, human factors, and systemic issues. [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1002/rob.70305)]

TABLE 2 | Standards and directives relevant to safety in haircutting robots.

Standard/Directive	Scope and content	Implications for haircutting robots
ISO/TS 15066 (International Organization for Standardization 2016c)	Defines collaborative operation modes (power & force limiting, hand guiding, speed & separation monitoring) and specifies pain threshold limits for different human body regions in collaborative robotics	Cutting tools must respect force/pressure thresholds for sensitive regions (e.g., ear, scalp). Motion planning requires separation distance or compliant actuation near the head.
ISO 13482 (International Organization for Standardization 2014)	Provides safety requirements for personal care robots, including mobile servant, physical assistant, and person-carrying robots	Haircutting robots fall under “personal care” class; design must ensure no hazardous contact, provide safe stopping functions, and integrate protective measures against entrapment or collision.
ISO 10218 (International Organization for Standardization 2025, 2011)	Specifies safety requirements for industrial robots and robot systems, including safeguarding, emergency stop, and collaborative operation principles	Although focused on industrial robots, safety provisions such as emergency stop, safeguarded space, and operator training can be adapted for haircutting robots.
ISO 18646 (International Organization for Standardization 2016a, 2016b)	Establishes performance and safety evaluation requirements for service robots, including navigation, obstacle avoidance, and functional tests	Relevant to haircutting robots as a service robotics category; requires standardized testing of positioning accuracy, safety during human contact, and robustness against unexpected user motion.
IEC 60601 (International Electrotechnical Commission 2005)	Establishes general safety principles for medical electrical equipment, including requirements for leakage current, thermal safety, and patient protection	Heating devices (e.g., dryers, curlers) integrated into haircutting robots must comply with electrical and thermal safety limits, preventing burns or electrical hazards.
IEC 80601 (International Electrotechnical Commission 2017)	Provides collateral and particular requirements for medical electrical equipment used in home healthcare environments	Haircutting robots intended for domestic use must ensure safe integration into non-professional settings, addressing risks from untrained users and variable home conditions.
ISO 14971 (International Organization for Standardization 2019)	Defines a systematic process for risk management of medical devices, including hazard identification, risk estimation, evaluation, and control	A structured risk analysis process can be applied to haircutting robots to document hazards, establish mitigation strategies, and justify acceptable residual risk.
EU Machinery Directive 2006/42/EC (European Union 2006)	Provides mandatory requirements for machinery placed on the EU market, covering design principles, documentation, and conformity assessment	Haircutting robots must undergo CE marking, ensure compliance with essential safety requirements, and address liability allocation in case of injury.

heightened user sensitivity in head interactions. These distinctive conditions underline the necessity of a tailored safety framework for haircutting robots.

3 | Risk Landscape and System Workflow

3.1 | Workflow Analysis and Risk Categorization

The robotic haircutting process can be structured into four successive stages: perception, planning, execution, and

feedback. Each stage involves critical operations but also introduces potential safety risks, as illustrated in Figure 2. Understanding this workflow is essential for identifying where failures may occur and how they propagate across the system.

In the **perception** stage, the robot acquires geometric and semantic information about the head and hair through multi-modal sensors. Errors in segmentation, calibration drift, or occlusions may lead to inaccurate recognition of sensitive regions such as the ears or scalp. If not corrected, these perception errors directly compromise subsequent planning.

During the **planning** stage, hairstyle models are combined with safety constraints to generate feasible cutting trajectories. Risks arise when trajectories are computed too close to the skin, when human motion is not adequately predicted, or when algorithmic errors bypass safety margins. Additional vulnerabilities occur in teleoperated settings, where network latency or cyberattacks may modify or delay trajectory commands.

In the **execution** stage, the robotic manipulator operates clippers, scissors, or heating tools in direct contact with the user. Physical hazards include excessive contact forces, unsafe blade geometry, overheating of dryers or curlers, and unintended collisions between the robotic arm and the head. These risks are critical because they involve immediate physical interaction with the user.

Finally, the **feedback** stage integrates sensor monitoring and post-operation assessment. Failures in this stage, such as delayed anomaly detection or incomplete task confirmation, may allow hazardous actions to continue without correction. Inadequate feedback also prevents proper adaptation and learning for subsequent tasks.

Overall, risks can be grouped into five overarching categories: network and cybersecurity, perception and sensing, physical and tool-related, control and algorithmic, and human-related. Table 3 consolidates these categories by aligning typical issues in haircutting scenarios with the associated system components. This categorization reveals that risks are not confined to a single function but rather distributed across communication infrastructure, sensing hardware, physical tools, control algorithms, and human interaction. The presence of physical risks stemming from tools and actuators emphasizes the role of inherent safety design. Perception-related vulnerabilities highlight the importance of redundant sensing and monitoring. Control instabilities demonstrate the necessity of robust decision-making and supervisory mechanisms.

Based on this analysis, several key questions arise during the haircutting process:

1. How can the robot accurately detect the detailed geometry of the human head and the dynamic environment?

2. Once this information is obtained, how can the robot execute the required actions safely?
3. If an unexpected event occurs, how can injury to the user be reduced or avoided?

To address these challenges, a three-layer safety framework is proposed to systematically mitigate risks in haircutting robotic systems, as illustrated in Figure 3. The **mechanical layer** provides baseline protection through compliant design, guarded tool geometry, and safe material selection, ensuring that unintended physical contact does not result in severe harm. The **sensing layer** integrates multimodal sensing and redundancy, enabling reliable detection of head geometry, user motion, and abnormal interactions. The **algorithmic layer** governs motion planning, controller stability, and cybersecurity defenses, supporting safer task execution under uncertainty or system delays.

These layers are interconnected rather than independent. The mechanical layer establishes the physical boundaries within which perception operates reliably. The sensing layer supplies real-time information that constrains and guides algorithmic decisions. In turn, the algorithmic layer supervises both mechanical actuation and sensing feedback, enforcing safety policies and corrective actions. Through this hierarchical and interdependent structure, safety responsibilities are distributed across hardware, sensing, and control levels, reducing the risk of systemic failure and avoiding reliance on any single safeguard. Sections 4–6 provide detailed discussions of each layer.

4 | Mechanical Layer

The mechanical layer forms the foundation of the safety framework in robotic haircutting, as it defines the physical materials and structural geometry of the system. It provides the first line of defense for user safety by shaping external interfaces, embedding passive protections, and incorporating inherent compliance. Mechanical safety is particularly critical because any malfunction or design flaw at this layer cannot be fully compensated by higher-level sensing or control.

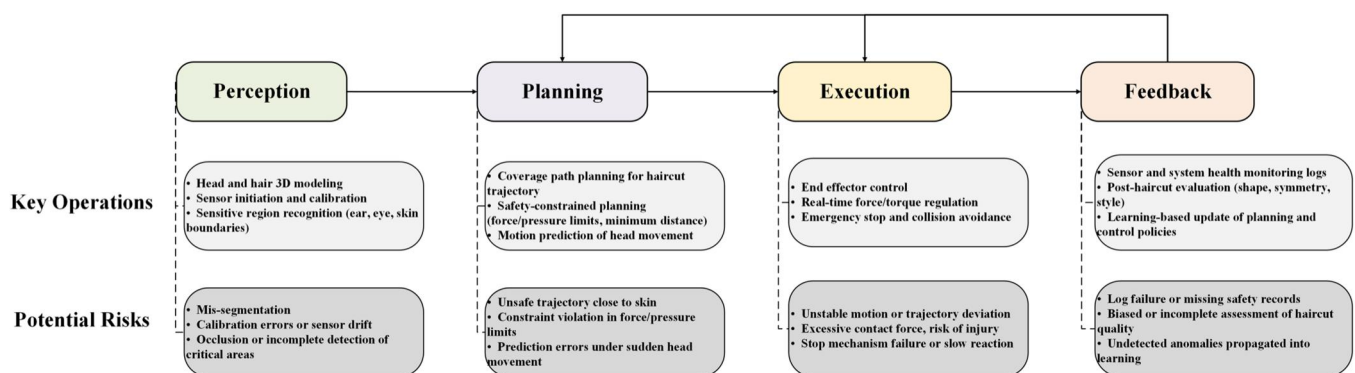


FIGURE 2 | Workflow stages of haircutting robots with key operations and potential risks. Each stage-perception, planning, execution, and feedback-includes representative tasks and associated safety risks, illustrating the progression from sensing to decision-making, action, and post-operation learning. [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1002/rob.70305)]

TABLE 3 | Categorization of risks in haircutting robots across system aspects.

Risk category	Typical issues	Associated components
Network & cybersecurity	• High latency in teleoperation causing delayed reaction • Network congestion leading to packet loss • Cyberattacks that hijack robot control • Unauthorized modification of haircut trajectory • Service interruption from unstable wireless connection	Communication link, wireless modules, teleoperation interface, network protocols
Perception & sensing	• Misclassification of hair, skin, or ear boundaries • Depth camera errors due to lighting or occlusion • Force/torque sensor drift or calibration failure • Loss of vibration or acoustic signals during tool-hair interaction • Single-sensor dependency without redundancy	Cameras, depth sensors, force/torque sensors, acoustic/vibration sensors, sensor fusion modules
Physical & tool-related	• Excessive contact force from clippers or scissors • Unsafe blade geometry causing skin injury • Overheating of dryers or curlers near scalp • Accidental collision with robot arm or end-effector • Tool wear leading to irregular performance	Clippers, scissors, heating tools (dryer/curler), blade geometry, robotic arm end-effector
Control & algorithmic	• Instability of controller under time delay • Unsafe motion planning without safety margins • Lack of fallback strategies when sensors fail • Runtime monitoring failure to stop hazardous action • Inconsistent tool switching logic	Motion planner, obstacle avoidance module, controller, runtime safety monitor, tool-switching logic
Human-related	• Sudden head movements (rotation, tilting, reflexes) • Postural shifts due to discomfort or fatigue • Fear or psychological stress during close contact • Non-cooperation or unexpected user actions • Lack of trust or acceptance of robotic haircutting	User head, posture, facial regions (ear/neck), behavioral response, psychological state

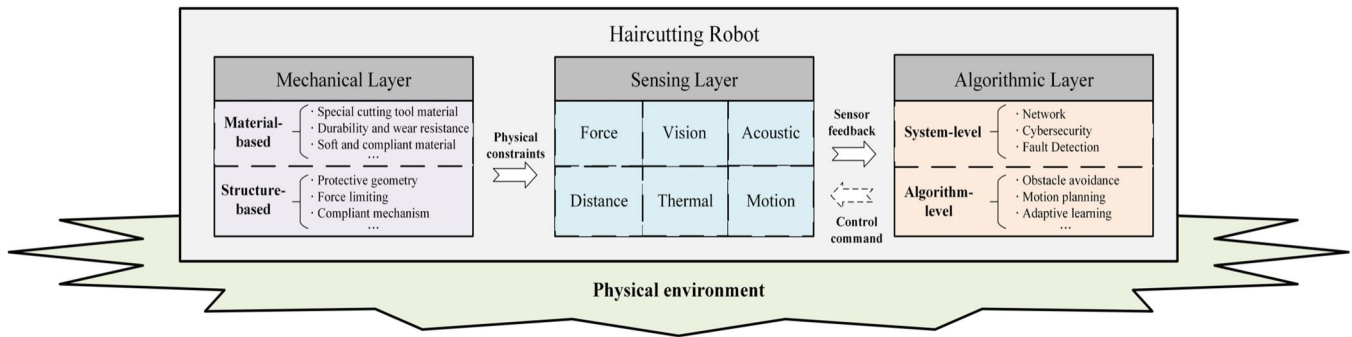


FIGURE 3 | Three-layer safety framework for haircutting robots with layer-wise components. [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com)]

TABLE 4 | Real-world mechanical incidents and issue analysis.

Instance	Description	Issue analysis
Instrument tip-cover over-advancement in robotic surgery (Singh et al. 2016)	Tip-cover accessory advanced too far, increasing scissor diameter and preventing smooth insertion/removal from cannula; no error alert	Assembly misalignment and poor tolerance design
Robotic surgical arm collisions triggering “power limit exceeded” (Rajih et al. 2017)	da Vinci system arms collided in certain cases, exceeding power output limits due to rigid structure and workspace overlap	Excessive arm rigidity and insufficient collision buffering
Review of da Vinci surgical system failures (Kim et al. 2009)	Among 1797 operations, 43 (2.4%) had malfunctions, including arm, optical, and instrument failures, often linked to wear, fatigue, or poor durability	Material fatigue and inadequate structural durability
Force/pressure thresholds in human-robot collisions (Han et al. 2024)	Experiments on pigskin and other tissues measured pain thresholds under impacts; highlighted role of shape, surface hardness, and edge design in injury severity	Hard surfaces and sharp edges lacking compliant materials

4.1 | Mapping from Real-World Incidents to Haircutting Scenarios

Analysis of real-world incidents summarized in Table 4 shows that failures at the mechanical layer often result in direct and severe consequences. Assembly misalignments, such as mispositioned accessories, demonstrate how even small tolerance errors can immediately compromise safety without leaving room for software correction. Arm collisions caused by excessive rigidity highlight that control responses are insufficient once unsafe forces have already been transmitted. Long-term fatigue and wear indicate that material degradation silently reduces reliability until sudden breakage occurs. Moreover, studies on force thresholds confirm that surface geometry and hardness are decisive factors for injury severity, regardless of force limits imposed by algorithms.

These findings emphasize the critical role of mechanical protection: once hardware design or materials are inadequate, sensing and control layers cannot fully prevent harm. For haircutting robots, the same patterns can be projected. Misassembly of clipper guards could expose blades, leading to accidental cuts. Rigid manipulators without compliance may cause bruises if the customer moves unexpectedly. Fatigue of tool heads or joints may result in cracks or detachment close to the scalp, creating sharp fragment hazards. Hard or sharp

housing surfaces could injure the skin even under minor contact.

Therefore, lessons from existing robotic systems indicate that the mechanical layer must provide inherent safety margins in both materials and structures. Without such safeguards, haircutting robots risk exposing users to direct physical harm, undermining both functional reliability and user trust.

4.2 | Safety Solutions

To mitigate these risks, safety can be enhanced through two complementary approaches: material-based design and structure-based design (see Figure 4).

4.2.1 | Material-Based Safety Design

- **Special cutting tool materials:** For example, ceramic blades are favored over conventional metallic blades because of their lower thermal conductivity, higher wear resistance, and reduced cutting aggressiveness (Jaiswal and Jaiswal 2024). Such material choice decreases the likelihood of accidental skin laceration while minimizing heat transfer to the scalp during prolonged operation.

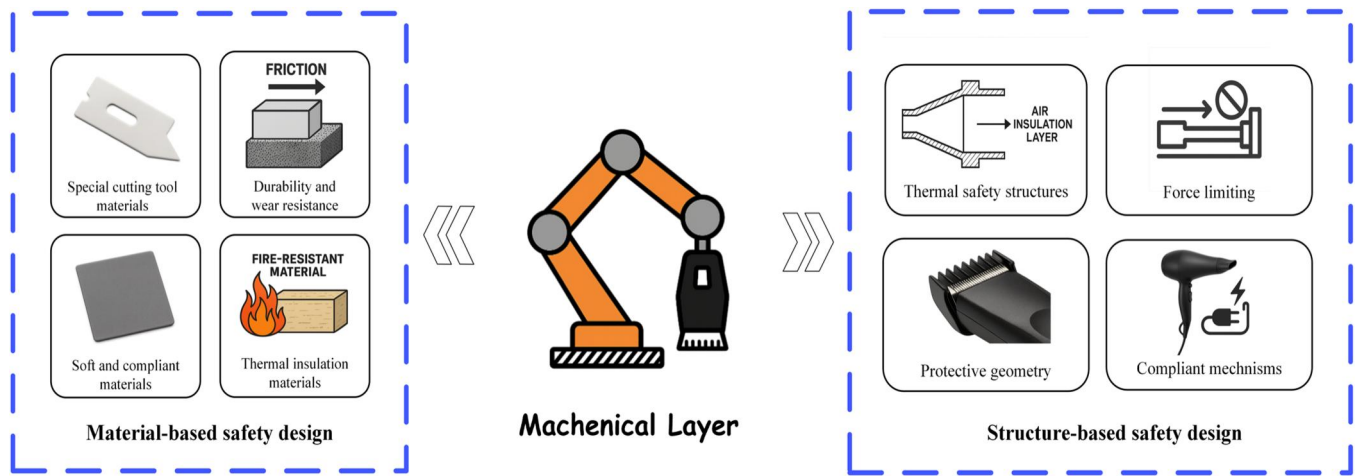


FIGURE 4 | Mechanical safety strategies categorized into material-based and structure-based designs. [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com)]

- **Soft and compliant materials:** Surrounding the housing with elastomeric layers (e.g., silicone or TPU, Shore A 20–40) allows localized impact energy to be absorbed and contact pressure to be distributed. According to ISO/TS 15066, the allowable transient contact force for the human head is approximately 65 N, with pressure thresholds around 90 N/cm² (Behrens et al. 2022); compliant layers help maintain contact within these safe ranges.
- **Thermal insulation materials:** The use of heat-resistant polymers and ceramic barriers, combined with double-wall housings containing air gaps, ensures thermal isolation of blow dryers or curling devices. These passive structures limit surface temperature rise below 43°C–45°C (Martin and Falder 2017), corresponding to safe exposure thresholds for human skin.
- **Durability and wear resistance:** Surface coatings such as diamond-like carbon (DLC) or polytetrafluoroethylene (PTFE) reduce blade friction, improve resistance to corrosion from hair care products, and extend service lifetime. Studies on hair clippers and blades have reported improved durability under cyclic use, although cutting-cycle counts vary with use conditions and testing methods. For example, DLC coatings have reduced volumetric wear by up to 75%–99% in simulated body/physiological environments (Madej et al. 2025).

4.2.2 | Structure-Based Safety Design

- **Protective geometry:** Protective geometry can be considered from two complementary perspectives. First, tool-level design focuses on the geometry of the cutting element itself, where rounded blade edges and increased edge radius reduce penetration risk compared with sharp tips below 10 μ m (Hainsworth et al. 2008), and blunt edge shaping further decreases the likelihood of laceration (Liu et al. 2021). Second, structural-level design introduces additional barriers such as distance combs, foil guards, or dual-layer structures (e.g., comb teeth plus transparent screens), which allow hair to enter the cutting zone while preventing skin intrusion (Krenik 2018). These complementary approaches create redundant safety boundaries

that passively protect the scalp even in cases of sudden user movement.

- **Force limiting mechanisms:** Mechanical devices such as slip clutches, torque fuses, or magnetic couplers are designed to disengage the drive once transmitted torque exceeds a threshold (Feizi et al. 2023; Zheng et al. 2021). This prevents excessive contact forces on the scalp in scenarios of tool jamming or abrupt external disturbance, offering an inherent safety margin independent of control response time.
- **Compliant mechanisms:** Series elastic couplings integrated at structural joints provide inherent compliance by absorbing impact forces (Robinson et al. 1999). Even if the robot contacts the user unexpectedly, the resulting interaction forces remain within tolerable ranges before higher-level intervention occurs (McCarthy et al. 2005).
- **Thermal safety structures:** Double-wall nozzles with integrated air insulation and passive thermal cut-off devices, such as bimetallic switches, act as fail-safe protections in heated styling modules. For example, early hair dryer designs included nozzles with inner and outer walls and an overload thermal switch placed within the nozzle structure, which interrupts heating when excessive temperatures are reached (Dreher 1976). Commercial bimetallic thermal switches (e.g. ST-12) are used in present-day hair dryers for passive temperature limiting (St-12 Bimetal Thermal Cutout Switch for Hair Dryer n.d). These mechanisms help ensure that the temperature at the scalp interface does not exceed safety thresholds even in the event of control malfunction or prolonged stall conditions.

Mechanical-layer safety designs directly shape the feasibility of future haircutting robots. Effective choices in materials and structural configurations not only reduce the likelihood of injuries but also enhance comfort, reliability, and user acceptance in close-contact tasks around the head. By embedding inherent safety into the hardware itself, designers can lower the reliance on complex sensing and control safeguards, creating systems that are more robust to uncertainties in human motion and environmental variability. This provides a practical

pathway toward deploying haircutting robots in everyday service contexts with both functional reliability and public trust.

5 | Sensing Layer

Building on the inherent protections of the mechanical layer, the sensing layer extends safety into the domain of perception. While structural design constrains physical risks, sensors determine whether the robot can adapt to human variability and environmental uncertainty in real time. In haircutting tasks, where close contact with the head is inevitable, the ability to reliably sense force, motion, distance, and temperature forms the foundation for proactive risk management. This section focuses on how sensing systems contribute to safety by transforming raw measurements into reliable indicators of the interaction state.

5.1 | Mapping from Real-World Incidents to Haircutting Scenarios

Analysis of the incidents summarized in Table 5 indicates that sensor shortcomings can rapidly translate into unsafe behavior. First, insufficient sensing modalities and absent fusion safeguards allow critical misidentifications, such as a human classified as cargo, indicating that single sensing streams are brittle to appearance variation. Second, single-sensor dependency without redundancy or cross-validation (e.g., faulty angle of attack (AOA) input) can drive the system into hazardous states even when all downstream logic operates nominally. Third, unhandled occlusion and lack of fallback sensing demonstrate that perception can collapse to unusable states after minor disturbances. Fourth, limited low-light capability and inadequate calibration degrade detection timeliness and confidence precisely when fast reaction is needed. Finally, optical degradation without real-time compensation (such as fogging, soiling, and defocus) reduces effective visibility and silently erodes safety margins.

When mapped to haircutting robots, the implications are direct. Relying on a single RGB camera risks ear/skin misclassification under hairstyle and complexion diversity; therefore, a fusion design with depth, proximity, or force cues is required to bound tool-skin clearance. Any single-point sensor failure (e.g., a sole proximity sensor drifting) could remove contact guards and permit unsafe approach; dual-channel sensing with cross-checks should be mandated for clearance and force. Occlusion events, including hair tufts, hands, or accessories blocking the lens, must be detected and handled with fallback proximity/force supervision to avoid blind operation. Low-light or miscalibrated cameras can delay boundary detection near the temple and nape; on-device illumination, periodic self-calibration, and confidence-aware gating are needed to prevent late reactions. Environmental optical degradation (spray mist, steam, or sebum) should trigger cleaning prompts or automatic derating, while thermal and vibration side channels monitor for emerging hazards. Therefore, these mappings show that sensor-layer safety in haircutting robots hinges on modality diversity, online health monitoring, and explicit fail-safe behaviors whenever sensing confidence drops.

5.2 | Safety Solutions

Sensing devices form the backbone of safety assurance in haircutting robots, as they provide real-time information about contact forces, motion dynamics, spatial boundaries, temperature, and tool conditions. These signals enable the robot to enforce limits on interaction, detect abnormal events, and maintain safe operating conditions.

5.2.1 | Safety Assurance by Individual Sensors

At the sensing layer, devices can be categorized by the type of data they provide, as each modality contributes specific mechanisms for hazard prevention:

Force and current sensing supports safe physical interaction. Motor current signals allow indirect estimation of joint torques

TABLE 5 | Real-world sensor-layer incidents and issue analysis.

Instance	Description	Issue analysis
Industrial robot in Korea misidentifies worker as a box (The Economic Times 2023)	Robot compressed an inspector after classifying him as cargo	Insufficient sensing modalities and lack of sensor-fusion safeguards
Boeing 737 MAX crashes due to faulty AOA sensor (BBC News 2019)	Angle-of-attack sensor error triggered inappropriate nose-down commands; two crashes	Single-sensor dependency without redundancy or cross-validation algorithms
Waymo car with occluded sensors halts abnormally (The Verge 2018)	Post-minor-incident tape covered LiDAR/cameras; environment not perceived correctly	No fallback sensing or occlusion-detection mechanisms
Uber AV fails to recognize pedestrian, fatality (The New York Times 2018)	System failed to timely detect a pedestrian walking a bicycle, no effective braking	Limited low-light sensing capability and inadequate sensor calibration
da Vinci endoscope fogging/focus loss reported in MAUDE (U.S. FDA n.d)	Endoscopic lens fogging or focus issues reduced visibility, contributing to errors	Optical sensor degradation under environment without real-time compensation

(Zhao and Nelson 2015), while six-axis force/torque sensors, strain-gauge load cells, piezoresistive pads, and capacitive or optical skins provide direct contact measurements (Lu et al. 2022). In haircutting robots, these measurements ensure that pressing forces remain within tolerable limits. For instance, when a clipper guard becomes misaligned and exposes the blade, abnormal torque or local contact pressure can be detected and used to trigger immediate motor shutdown. Distributed tactile pads further localize contact points around sensitive regions such as the ear or temple, where excessive force could otherwise result in bruising or laceration (Kappassov et al. 2019).

Motion and acceleration sensing enables rapid collision detection. Inertial measurement units (IMUs), consisting of accelerometers and gyroscopes, capture sudden kinematic changes: high-frequency accelerometers detect sharp impacts, while gyroscopes identify unexpected angular deviations (Ionut-Cristian and Dan-Marius 2021). In haircutting robots, these signals are essential for handling unpredictable head motion by the customer. For example, if the head rotates suddenly while the robot is trimming near the ear, accelerometer spikes can trigger an emergency stop, while gyroscope readings allow trajectory adjustment to prevent scraping. The short response time minimizes exposure to hazardous contact (Borowska-Terka and Strumillo 2023).

Vision and distance sensing provide spatial awareness. RGB and stereo cameras, structured-light, and time-of-flight (ToF) systems detect boundaries between hair, skin, and ears (Mur-Artal and Tardós 2017; Tsuji and Kohama 2019; Valkenburg and McIvor 1998). For near-field safety, infrared, ultrasonic, or optical reflectance sensors enforce minimum tool-scalp distances. In haircutting robots, this dual-layer perception helps ensure that even if hair partially occludes the visual field, proximity sensors can still help prevent cutting blades from directly touching the scalp (Fonseca et al. 2022). Furthermore, vision-based algorithms can identify the contour of ears and prevent unintended clipping injuries, a common source of discomfort in manual haircuts.

Thermal sensing safeguards heated tools. Thermistors, thermocouples, resistance temperature detectors (RTDs), and infrared thermopiles measure point temperatures, while thermal cameras or multi-pixel arrays capture broader temperature distributions (Wade et al. 2017; Afzal et al. 2022; Coşar et al. 2018). In haircutting robots that integrate blow dryers or curling devices, these sensors enforce automatic power reduction or cutoff when surface temperatures approach human tolerance levels. This prevents scalp burns in scenarios where airflow is blocked by hair or the tool is held too close to the skin for prolonged periods. Thermal imaging can further detect localized hot spots, providing redundancy against failures of single-point sensors.

Vibration and acoustic sensing supports the detection of anomalies that precede mechanical hazards. Micro-Electro-Mechanical Systems (MEMS) accelerometers, piezoelectric sensors, and microphone arrays identify abnormal oscillations, grinding sounds, or resonance shifts caused by blade wear, imbalance, or entanglement with hair (Li et al. 2022; Zhang

et al. 2022). In haircutting robots, such early warnings are critical to prevent escalation into dangerous events. For instance, if hair entangles within the blade assembly, the acoustic signature changes sharply, detecting this pattern enables the system to disengage the motor before the jam translates into pulling, skin pinching, or fragment release (Wang et al. 2020).

However, each modality has inherent limitations. Vision can be affected by hair occlusion or illumination, force sensors add cost and weight, IMUs are sensitive to noise, thermal sensors may respond too slowly, and vibration and acoustic sensing are prone to background interference. In the complex environment of haircutting, which is characterized by fine hair structures, reflective surfaces, humidity, and unpredictable head motion, relying on a single sensor type is insufficient.

5.2.2 | Redundancy and Multimodal Integration

Although single sensors provide essential functions, haircutting robots must operate under variable lighting, hair occlusion, humidity, and unpredictable motion. Robust perception therefore depends on redundancy and multimodal fusion.

Sensor redundancy ensures fault tolerance by monitoring critical variables through overlapping modalities. For example, scalp clearance can be verified simultaneously by proximity sensors and cameras, while both current-based torque estimation and direct force sensing offer independent measures of contact load (Masalskyi et al. 2025). This redundancy helps maintain the observability of safety-critical variables even if one sensor is obstructed by hair, degraded by cosmetic residues, or temporarily malfunctioning (Saleem et al. 2025).

Multimodal fusion enhances robustness by combining data from different sensing sources (see Figure 5). Fusing IMU data with acoustic features enables rapid detection of tool jamming, while integrating vision with force sensing distinguishes intentional hair trimming from accidental skin contact (Khattak et al. 2018). Such fusion compensates for environmental variability, reduces the need for intrusive sensor placement, and aligns perception outputs with established human safety thresholds (Han et al. 2025). By leveraging complementary information, multimodal sensing provides a more reliable basis for initiating protective actions.

Overall, redundancy and multimodal fusion transform isolated sensing modalities into a coherent perception system. By continuously constraining forces, monitoring motion, enforcing clearances, regulating temperature, and detecting anomalies, the sensing layer enables haircutting robots to identify and mitigate risks in real time.

The sensing layer thus bridges the gap between physical safeguards and algorithmic control. By consolidating heterogeneous signals into consistent safety indicators, it provides the information basis for higher-level planning and supervisory mechanisms. In this way, sensing information does not replace mechanical design but complements it, ensuring that safety is actively monitored and communicated to the control layer. The next section builds on this foundation by examining how control strategies enforce safe execution under uncertainty.

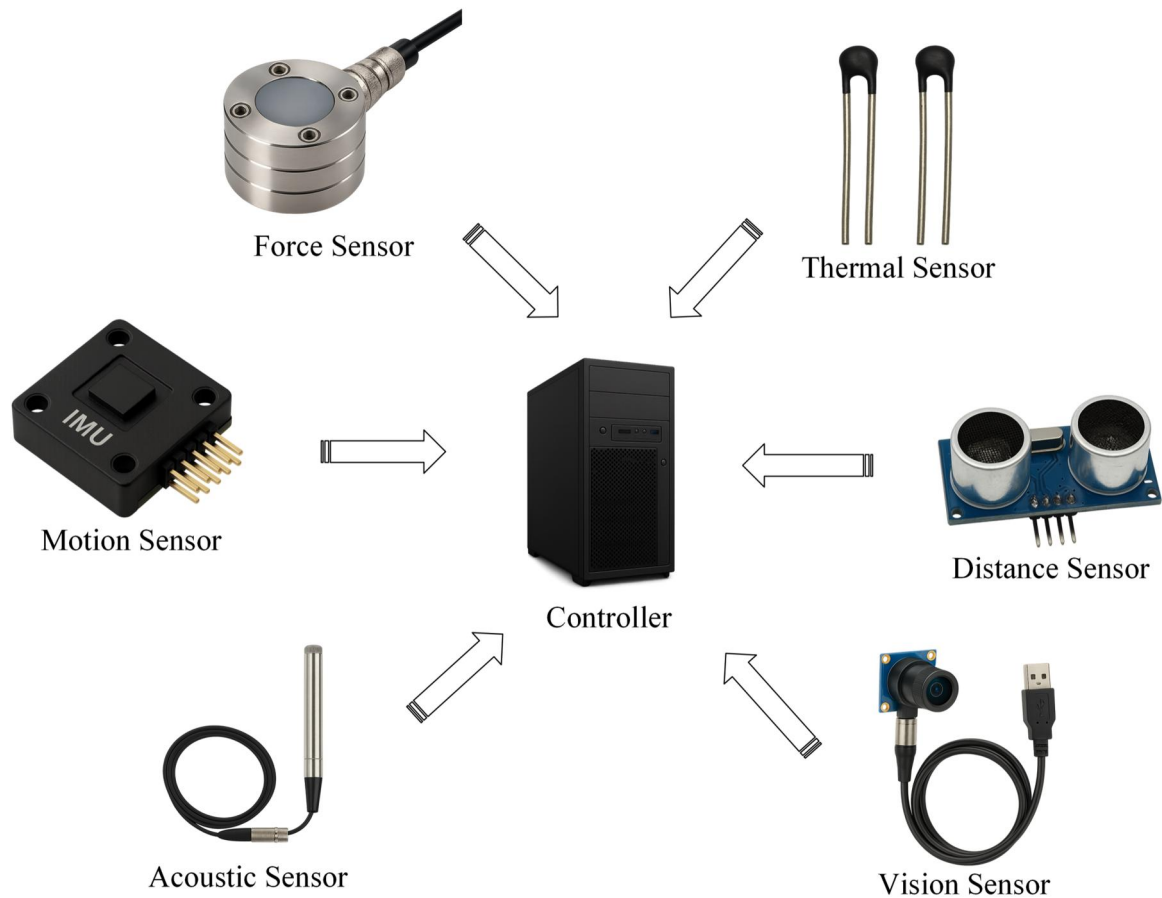


FIGURE 5 | Multimodal sensor fusion framework for haircutting robots, enabling complementary information to be combined for robust perception. [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1002/rob.70305)]

6 | Algorithmic Layer

The mechanical and sensing layers provide fundamental hardware safeguards and perceptual capabilities, while the algorithmic layer is ultimately responsible for transforming raw signals and structural constraints into safe and coordinated actions. It governs motion planning, trajectory generation, real-time control, and supervisory logic, thereby serving as the decisive interface between human intent, sensory perception, and physical execution.

6.1 | Real-World Incidents and Mapping to Haircutting Robots

Real-world incidents across autonomous driving, surgical robotics, and teleoperation reveal recurring vulnerabilities within the algorithmic and control layer (see Table 6). A recurring pattern is that algorithmic errors or missing safeguards, rather than hardware malfunctions, often act as the root cause of safety-critical failures. For instance, crashes involving autonomous vehicles such as the Uber ATG Tempe case and Tesla collisions with stationary objects, demonstrate how incomplete perception-prediction pipelines or biased collision-avoidance logic can lead to severe consequences even when mechanical subsystems remain intact. Similar issues have been documented in medical robotics, where recalls of surgical platforms highlighted unexpected actuator motions triggered by

software anomalies, underscoring the lack of built-in fail-safe routines. Similarly, studies on telesurgery and teleoperated robots have shown that large network delays or deliberate cyberattacks can directly destabilize control, inducing unsafe end-effector motions or preventing timely corrective actions.

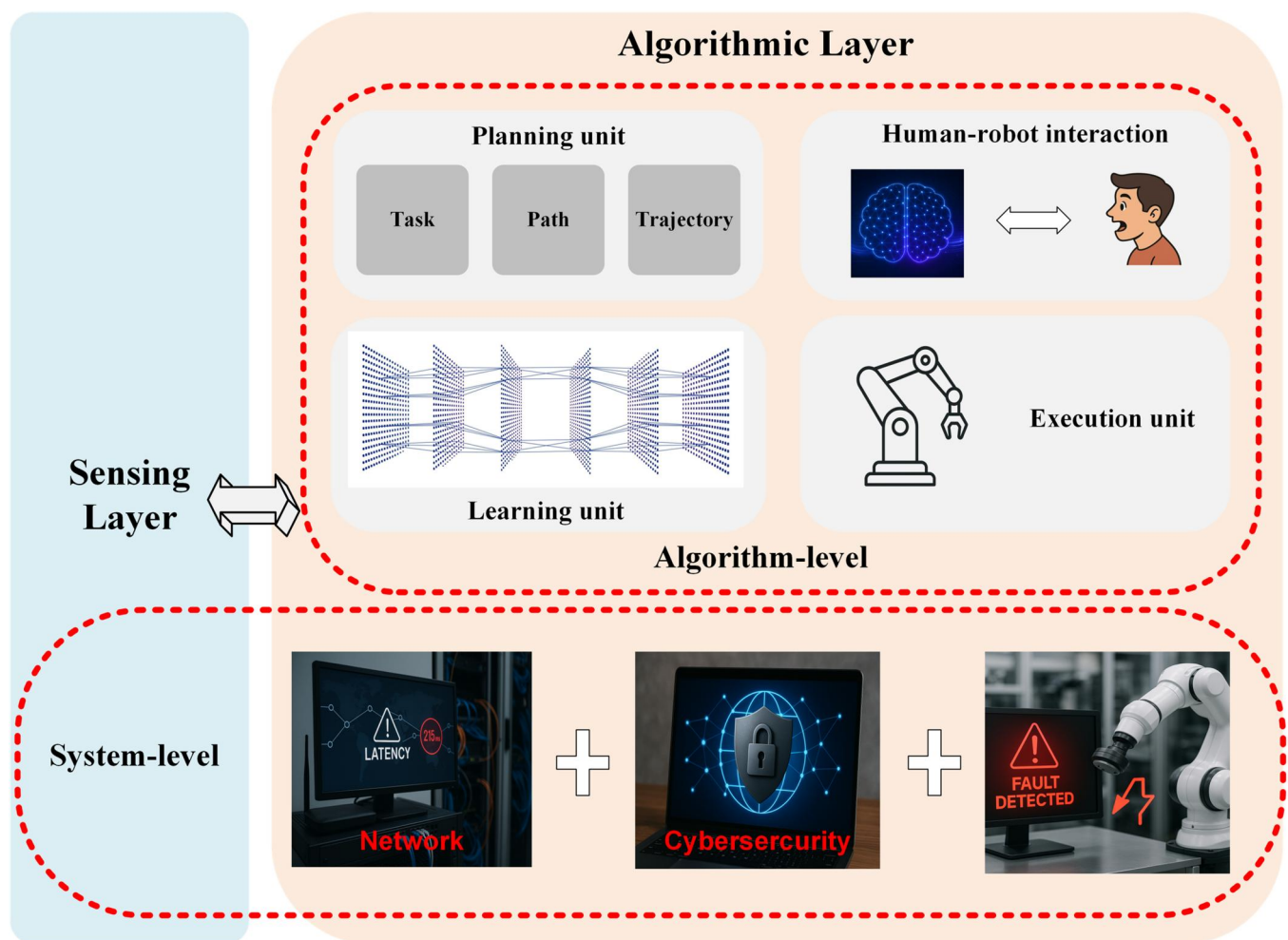
When mapped to haircutting robots, these observations indicate that algorithmic fragility could readily translate into hazardous behavior in close-contact human environments. A failure to reliably predict head motion or to maintain skin-tool clearance could expose the scalp or ears to laceration. Software exceptions without supervisory override may cause unintended blade movement near sensitive regions. Latency in remote operation or cloud-based processing could delay responses to sudden user motion, resulting in unsafe contact before correction is possible. Moreover, compromised communication channels could enable malicious manipulation of motion commands, threatening both user safety and trust. These scenarios underline that, in haircutting robots, the algorithmic layer must be designed not only for nominal performance but also for resilience against perception errors, software anomalies, and degraded or adversarial network conditions.

6.2 | Safety Solutions

As shown in Figure 6, safety mechanisms at the algorithmic layer operate at two complementary levels. First, supervisory

TABLE 6 | Real-world algorithm/control-layer incidents and issue analysis.

Instance	Description	Issue analysis
Uber ATG Tempe fatal crash (NTSB 2019)	NTSB found the ADS failed to correctly classify/predict the pedestrian and emergency braking was disabled in autonomy, relying on a distracted safety driver	Incomplete perception-prediction algorithm and disabled safety override logic
Tesla Autopilot crashes into stationary vehicles (EA22-002) (The Verge 2025)	NHTSA aggregated > 200 frontal-plane crashes, including first-responder scenes with stationary obstacles, under driver-assist engagement	Algorithmic bias toward moving objects; inadequate collision-avoidance software
FDA Class II recall: da Vinci Xi software anomaly (U.S. FDA 2017)	FDA notice reported that a software anomaly could cause unexpected master movement and unintended instrument tip motion	Software defect in control logic without fail-safe routines
Teleoperated surgical robot cyberattacks (Raven II) (University of Washington News 2015)	Researchers demonstrated that packet manipulation/DoS attacks could induce unsafe motions or cause command loss in a teleoperated surgery robot	Lack of secure communication protocols and software-level integrity checks
Telesurgery delay limits and performance degradation (Quist 2023)	Reports indicated that large delays (≥ 500 –750 ms) sharply increase errors and can render direct teleoperation unsafe	Control algorithms not robust to high-latency networks
Tesla Autopilot misclassifies white trailer as sky (Reuters 2016)	Autopilot failed to detect a white semi-trailer crossing the road; no braking, fatal crash	Faulty perception algorithm and misconfigured sensor-fusion filters

**FIGURE 6** | Cross-layer safety unit for haircutting robots. The unit integrates network monitoring, cybersecurity, and fault detection as supervisory modules that operate across both sensing and algorithmic processes, ensuring robustness against latency, malicious interference, and hardware failures. [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1002/rob.70305)]

mechanisms for network reliability, cybersecurity, and runtime fault detection span perception and control to ensure resilience against timing uncertainties, adversarial interference, and internal faults. Second, safety can be embedded throughout the algorithmic stack, including state estimation and uncertainty quantification, human-motion prediction and intent recognition, decision making and task scheduling, motion planning and control, and data-driven adaptation under explicit safety constraints. In this way, unsafe actions can be reduced throughout the haircutting workflow.

6.2.1 | System-Level Reliability and Protection

Network reliability. In teleoperated haircutting scenarios, data transmission latency or congestion can undermine stability and operator control (Sheridan 2002). A two-stage strategy is essential. Before a session begins, the system evaluates round-trip delay and available bandwidth, authorizing operation only when latency is acceptable. During the haircut, continuous monitoring of quality of service ensures that if latency exceeds limits or packet loss increases, the robot either pauses or prompts for local human takeover. From a control perspective, predictive and passivity-based controllers can mitigate delay-induced instability, while control barrier functions (CBFs) enforce safety envelopes around the scalp even under delayed actuation (Periotto et al. 2024; Uyulan 2024). Safe reinforcement learning further constrains policy exploration to reduce the likelihood that generated trajectories violate safety limits (Risiglione et al. 2021; Periotto 2023).

Cybersecurity. Both teleoperated and autonomous modes are vulnerable to cyberattacks. Encrypted communication, secure authentication, and integrity checks are therefore mandatory (Yaacoub et al. 2022; Iqbal et al. 2019). In addition, haircut-specific path-integrity safeguards can be employed: planned trajectories are generated and reviewed prior to execution, and during operation the executed path is continuously compared with the reference. If deviations exceed tolerance, whether due to hacking, operator error, or unforeseen anomalies, the robot halts or requests explicit confirmation before proceeding (Alemzadeh et al. 2016).

Fault detection and predictive maintenance. Even with secure communication and stable control, internal faults may propagate into unsafe behavior if undetected. Real-time monitoring of motor currents, vibration, and proximity data allows immediate detection of excessive forces, abnormal oscillations, or sensor degradation, prompting an emergency stop (Sobhi et al. 2023). Beyond reactive protection, predictive maintenance methods, such as anomaly detection, statistical health monitoring, and model-based prognostics, anticipate wear, backlash, or drift before they compromise safety, enabling timely intervention (Eang and Lee 2024; Zhang et al. 2025; Raouf et al. 2022).

6.2.2 | Algorithm-Level Control

In addition to module-level safety mechanisms, safety can also be embedded across the algorithmic stack, so that perception, decision-making, and control all operate within protective boundaries:

Obstacle avoidance. A common formulation is to enforce a minimum clearance between the tool and restricted head regions (scalp/ear/neck):

$$d(\mathbf{q}) \geq d_{\min},$$

where $d(\mathbf{q})$ is the signed distance from the tool geometry to the head model under robot configuration $\mathbf{q}$. This constraint can be integrated into sampling-based planning (e.g., rapidly-exploring random tree, RRT (Karaman et al. 2011)/probabilistic roadmap, PRM (Bohlin and Kavraki 2000)) by rejecting states violating $d(\mathbf{q}) < d_{\min}$, or into trajectory optimization by adding inequality constraints (Khatib 1986; Borenstein and Koren 1989; Pandey et al. 2017). Such distance-based safety constraints and their determination principles are also consistent with industrial robot safety standards.

Predictive motion planning. By forecasting head movement from vision or inertial cues, the robot can adjust tool trajectories proactively rather than reactively, reducing the likelihood of abrupt emergency stops and enabling smoother, safer interaction (Latombe 2012; Elbanhawi and Simic 2014; Canny 1988). In practice, such forecasts can be incorporated into the system model or treated as exogenous inputs within the planning horizon. A representative and widely adopted realization of this idea is model predictive control (MPC) (Mayne et al. 2000), in which future system behavior is optimized over a finite prediction horizon while explicitly enforcing safety constraints. Generally, the robot solves

$$\begin{aligned} \min_{\mathbf{u}_{0:N-1}} \quad & \sum_{k=0}^{N-1} \ell(\mathbf{x}_k, \mathbf{u}_k) + \ell_f(\mathbf{x}_N) \\ \text{s.t.} \quad & \mathbf{x}_{k+1} = f(\mathbf{x}_k, \mathbf{u}_k), \\ & (\mathbf{x}_k, \mathbf{u}_k) \in \mathcal{Z}, k = 0, \dots, N-1, \end{aligned} \quad (1)$$

where $\mathbf{x}_k$ denotes the system state, $\mathbf{u}_k$ the control input, $\ell(\cdot)$ and $\ell_f(\cdot)$ are stage and terminal cost functions, and $\mathcal{Z}$ represents the admissible set of state-input pairs. For robotic haircutting, the admissible set $\mathcal{Z}$ can be instantiated to encode task-specific safety and operational constraints. In particular, $\mathcal{Z}$ may include geometric clearance requirements, as well as interaction constraints that limit contact forces or velocities during incidental contact. These quantities can be incorporated via force estimates, impedance models, or conservative proxies commonly used in contact-aware planning.

By enforcing these constraints over a receding prediction horizon, MPC enables anticipatory motion adaptation to head movement while reducing reliance on abrupt emergency stops, thereby supporting smoother and more predictable interaction, which is particularly desirable in safety-critical haircutting scenarios.

Human-robot interaction. In close-contact haircutting, interaction safety increasingly relies on data-driven intent inference and uncertainty-aware decision making rather than explicit command-based control (Dragan et al. 2013). Learning-based perception models estimate head pose dynamics, micro-movements, and interaction cues from multimodal sensory inputs (Hempel et al. 2022), enabling the robot to anticipate user

behavior. Uncertainty-aware mechanisms, such as confidence estimation or predictive variance monitoring, allow the system to automatically reduce speed, enlarge safety margins, or request user confirmation when inference reliability decreases (Gal and Ghahramani 2016). In addition, motion-legibility and predictability methods explicitly shape robot motion to be easier for users to interpret, which has been shown to improve perceived safety, comfort, and trust in human-robot interaction, particularly around sensitive body regions.

Self learning and adaptation. Self learning enables hair-cutting robots to adapt to individual head geometry, hair properties, and user preferences through accumulated data rather than predefined rules (Argall et al. 2009). Representative approaches include learning-based perception and mapping of head geometry, user-specific motion prediction, and preference modeling from prior interactions (Jain et al. 2015; Christiano et al. 2017). In practical systems, self learning is typically performed offline or incrementally from low-risk data, while deployment-time behavior remains governed by fixed safety envelopes defined by geometric, force, and proximity constraints. Runtime monitoring and rule-based supervision ensure that self-learned models influence task-level planning and parameter tuning without directly overriding safety-critical control decisions (Seto et al. 1998). This separation allows personalization and gradual performance improvement while maintaining predictable and auditable safety behavior.

By combining supervisory safeguards with safety-aware algorithmic strategies, the control layer not only mitigates external disturbances and internal faults but also constrains the robot's decision-making and adaptation processes, supporting more predictable and safety-oriented operation throughout the hair-cutting workflow.

In conclusion, the algorithmic layer extends safety from passive protection toward proactive assurance. Supervisory modules protect against faults, adversarial influence, and communication uncertainties, while algorithmic strategies embed safety directly into perception, decision making, and control. Together, these mechanisms transform heterogeneous sensor inputs into explicit safety constraints and supervisory decisions, helping the robot respond more predictably to human variability and environmental disturbances. When combined with the mechanical and sensing layers, the algorithmic layer completes a multilayered framework in which safety is addressed not only by physical design and perceptual monitoring, but also by the logic that governs robot behavior.

7 | Experimental Proof-of-Concept (PPoC)

Because research on haircutting robots remains limited and few real robotic haircutting platforms are available, a series of PoC demonstrations are conducted to illustrate how the proposed three-layer safety framework can be applied in haircutting scenarios. It should be emphasized that these PoC demonstrations are not intended to validate safety performance or establish quantitative safety guarantees, but rather to illustrate how the proposed safety framework can be instantiated and decomposed across mechanical, sensing, and algorithmic layers.

7.1 | Experimental Setting

Real-world robotic haircutting is a highly complex scenario due to multiple factors, such as the diversity of tools, dynamic changes in the environment, and hairstyle esthetics. Figure 7 illustrates the overall workflow of a robotic haircutting task. During execution, the sensing layer and the algorithmic layer continuously monitor the task and provide feedback for decision-making and control, while the mechanical layer serves as the foundational protection to ensure basic operational safety. It is worth noting that different haircutting tools (e.g., hair clippers, scissors, razors) have different geometries and contact risks. Therefore, the settings in the sensing and algorithmic layers should be adjusted accordingly. In this paper, we present only a PoC demonstration using the simplest single-tool hair-clipper scenario. Our goal is to show how the proposed three-layer safety framework can be decomposed into distinct layers and how each layer contributes to safety.

The main experimental objectives are as follows:

- Based on the structural characteristics of a real hair clipper, analyze how the **mechanical layer** contributes to safety.
- Use the robot arm's built-in force estimation to monitor force variations when the clipper contacts the head model, and trigger a safe stop accordingly, demonstrating the safety feasibility of the **sensing layer**.
- Compare whether the clipper can reach a specified target position with and without obstacle-avoidance planning, highlighting the role of the **algorithmic layer** in avoiding restricted regions and ensuring reachability.

Hardware Setup: The experimental platform consists of a hair clipper equipped with a limit comb and a 7-DoF Kinova Gen3 collaborative robotic arm. All additional mounting and adapter components are fabricated via 3D printing. All experiments are performed on a statically fixed foam mannequin head to emulate a simplified haircutting scenario, no human subjects are involved in the experiments. The head model used in simulation is reconstructed from 3D point clouds captured using a Kreon articulated measuring arm with a scanning probe, followed by reverse engineering (see Figure 8). In this work, the coordinate-frame alignment between the simulated head model and the physical mannequin head is achieved via manual registration.

Software Setup: Robot control is implemented using the Python Kortex interface. Obstacle-avoidance planning in simulation is carried out in MATLAB R2025b, leveraging its native Kinova Gen3 components for modeling and validation.

7.2 | Demonstration for Mechanical Layer Safety

The mechanical layer provides tool-level passive safety protection that does not depend on perception or planning algorithms. In close-contact haircutting tasks, the main tool-related risks include accidental skin contact with sharp edges, pinching around the clipper head, and scraping during contact. For this PoC demonstration, we use a hair clipper equipped with a limit comb and a specialized cutting blade (see Figure 9) to illustrate

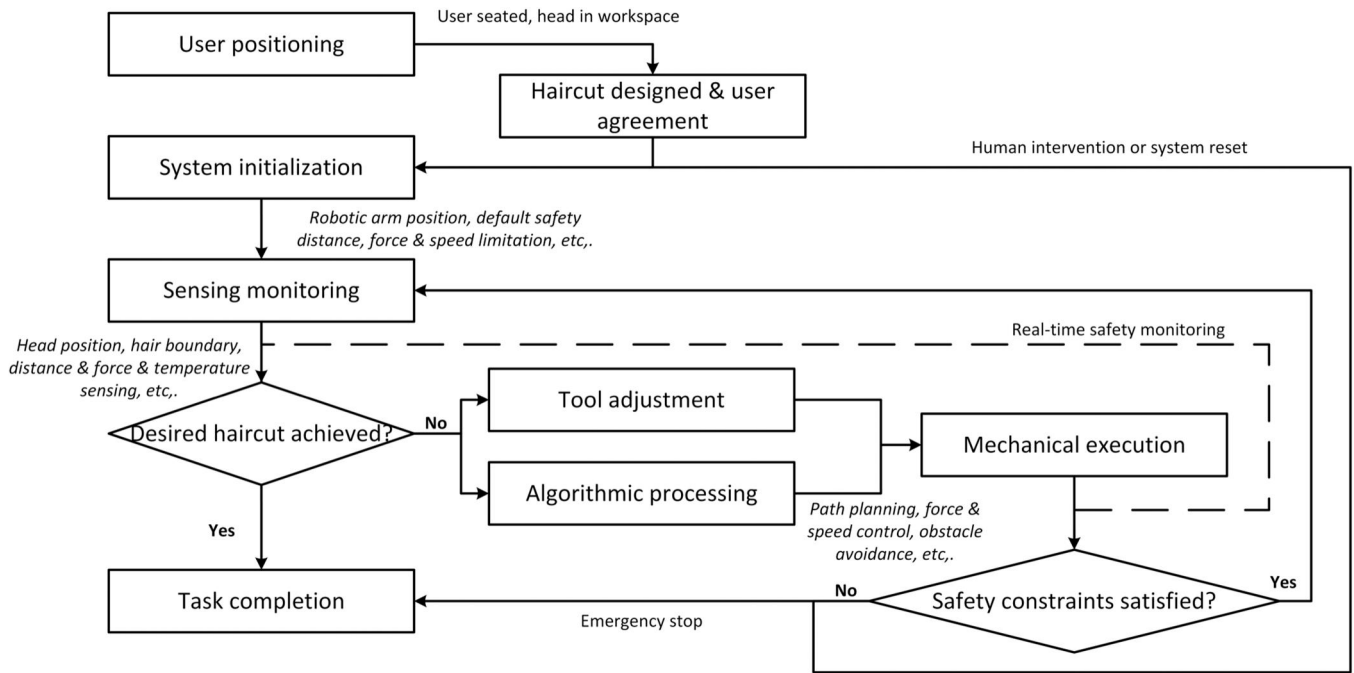


FIGURE 7 | Flowchart of the robotic haircutting process with a three-layer functional architecture. The sensing layer continuously monitors the entire process and provides rich multimodal information to the algorithmic layer, while the mechanical layer ensures underlying operational safety.

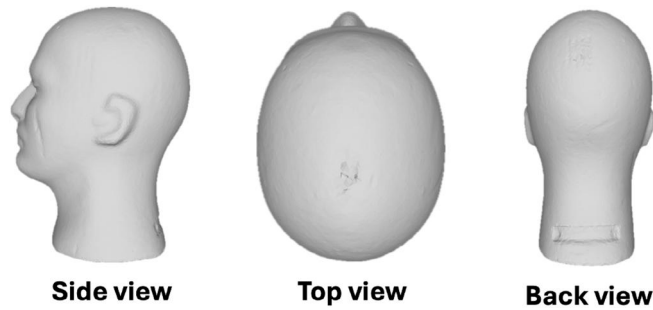


FIGURE 8 | 3D reconstruction of the human-head model.

mechanical-layer safety. Through the design of tool geometry and contact characteristics, the clipper reduces the likelihood of direct skin contact at the physical level and mitigates the severity of nicks or pinching during brief unintended contact.

Since these protections are inherent to the tool structure, they remain effective even when higher-level modules (e.g., perception or planning) produce errors. Therefore, the mechanical layer establishes a stable baseline safety boundary for the system, while the sensing layer and the algorithmic layer provide active monitoring and decision-making based on this foundation.

7.3 | Demonstration for Sensing Layer Safety

The primary role of the sensing layer is to continuously monitor risk-related signals during near-contact interactions and translate them into explicit and actionable safety triggers, allowing potentially hazardous actions to be interrupted before risks accumulate. In this PoC demonstration, we select the most

direct and contact-relevant signal, the internal end-effector force estimate provided by the Kinova Gen3 robot, to monitor the interaction between the hair clipper and the head model in real time (see Figure 10). When the monitored force exceeds a predefined threshold, a safety stop is immediately triggered to prevent further advancement and potential injury. As shown in Figure 10, during the approach and potential contact with the head model, the system continuously reads the robot's force-related estimation signals and applies a fixed force threshold of 10 N as the safety triggering condition. The estimated force components F_x , F_y , and F_z are obtained via the Python Kortex interface, and the final force magnitude is computed as $F_{\text{final}} = \sqrt{F_x^2 + F_y^2 + F_z^2}$. When the estimated force remains below the threshold, the robot is allowed to continue executing the planned motion; once the threshold is exceeded, a stop strategy is immediately activated, thereby realizing sensing-based safety protection.

During free-space motion and normal approach, the estimated force exhibits only bounded dynamic fluctuations. The triggering rule of the sensing layer tolerates such variations and does not interrupt task execution due to nominal motion-induced disturbances. As contact progresses, the estimated contact force at the end-effector increases rapidly and triggers the stop mechanism upon reaching the threshold, causing the robot to halt its forward motion. It should be noted that, at the moment of contact, the end-effector velocity is abruptly driven to zero by the rigid contact constraint, resulting in an impact effect. Since the end-effector force of the Kinova Gen3 is estimated from joint torque residuals, this impact is amplified into a sharp force peak under high stiffness and discrete-time control. Once motion stops, the contact force returns to within the threshold range, thereby preventing continuous force growth or repeated accumulation that could pose safety risks.

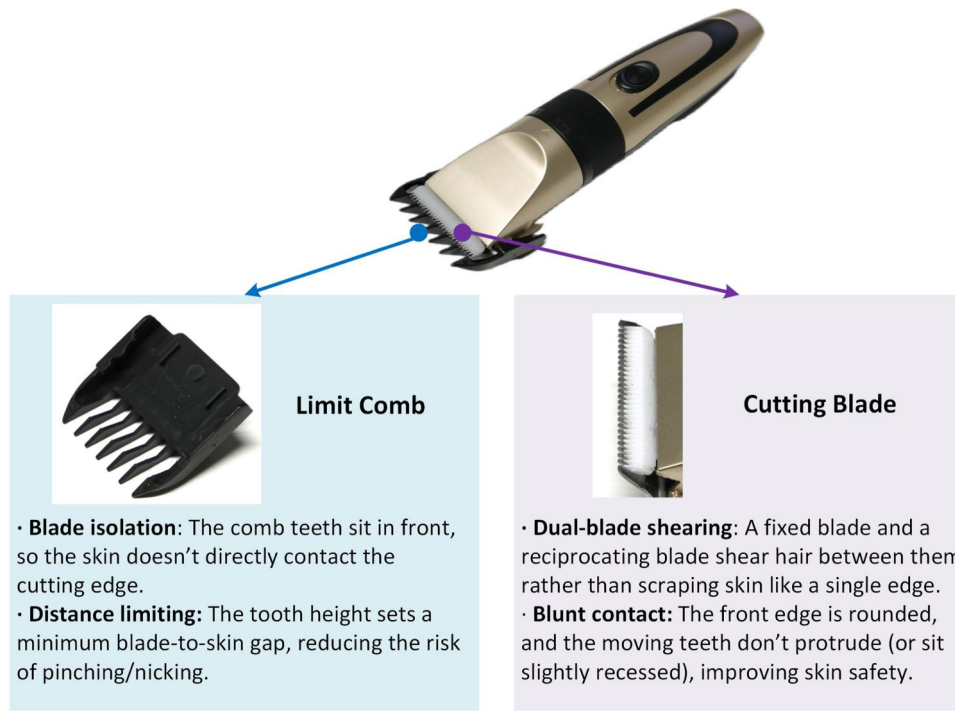


FIGURE 9 | Mechanical safety design of the hair clipper, with protection provided by the limit comb and specialized cutting blade. [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com)]

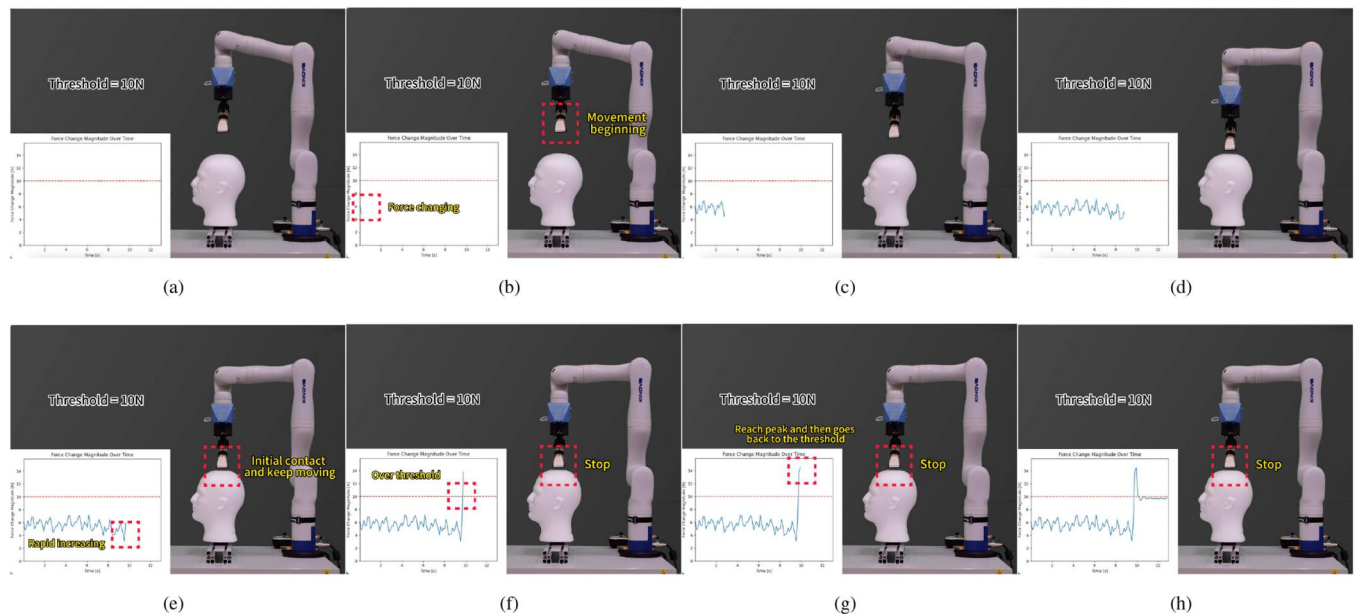


FIGURE 10 | Real-time monitoring of the estimated end-effector force with a predefined threshold for safety stop triggering. (a)–(e) show free-space motion and normal approach, where the force estimate exhibits minor fluctuations during motion initiation or posture adjustment but remains below the threshold. (f)–(h) depict the contact phase: when the hair clipper contacts the head model and advances along a constrained direction, the estimated force increases rapidly. Once the threshold is exceeded, motion is immediately stopped, preventing further force increase. [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com)]

In summary, the sensing layer provides a clear and implementable safety triggering mechanism through real-time monitoring of contact-related risk signals, enabling early detection and timely interruption of hazardous interactions. It is worth emphasizing that this mechanism operates on top of the tool-level baseline protection provided by the mechanical

layer. Moreover, the stop and risk-trigger signals generated by the sensing layer supply essential online information and safety boundaries for higher-level safety decision-making in the subsequent algorithmic layer, such as obstacle avoidance planning and reachability guarantees under constrained regions.

7.4 | Demonstration for Algorithmic Layer Safety

The algorithmic layer aims to generate constraint-satisfying motion decisions based on the environmental information and safety boundaries provided by the sensing layer, so that the system can maintain reachability and safety in the presence of restricted regions or obstacles. An initial point and a target point are defined for the hair clipper, and the head model and its surrounding area are treated as a restricted region. Two conditions are considered:

1. **Without obstacle avoidance:** the robot moves toward the target directly without explicitly considering the restricted region.
2. **With obstacle avoidance:** the algorithmic layer generates and executes a collision-free trajectory under the restricted-region constraints. The trajectory is first planned in MATLAB and then executed offline on the physical device.

This PoC demonstrates the role of the algorithmic layer when the hair clipper approaches a target near the head.

7.4.1 | Without Obstacle Avoidance

Without obstacle-avoidance planning, when the clipper moves toward the target, the motion can be blocked by the head geometry, preventing further advancement toward the target (the blocking case in Figure 11). To avoid risk accumulation caused by continued pushing in the blocked state, a force threshold was used as a protection rule: when the contact-related force signal exceeds the threshold, the robot stops. This result indicates that without algorithmic layer avoidance decisions, the system often ends in a reactive “blocking-stop” manner, and reachability cannot be guaranteed.

7.4.2 | With Obstacle Avoidance

In this case, obstacle avoidance is implemented using a non-linear model predictive control (NLMPC) framework provided by the MATLAB Model Predictive Control Toolbox. The overall structure of the planner follows an official example from the MathWorks documentation, which demonstrates collision-free trajectory generation for robotic manipulators using NLMPC in conjunction with the Robotics System Toolbox. The human head is modeled as a static obstacle in the workspace. A minimum safety distance is enforced between the robot and the head model through distance-based collision constraints, preventing the robot from entering restricted regions during motion execution. As shown in Figure 12, the simulation results

demonstrate that the robot reaches the target while maintaining a prescribed safety margin from the static head obstacle. Furthermore, the physical experiments illustrate that the Kinova Gen3 is able to follow the planned trajectory and reach the target point in this simplified setup, as shown in Figure 13. Notably, the simulation focuses on motion planning and obstacle avoidance at the robot level, without explicitly modeling the haircutting clipper geometry. The planning constraints are defined based on a minimum safety distance between the robot and the head model. Since the objective is to illustrate robot-level collision-free trajectory generation rather than validate tool-skin interaction safety, this simplification is acceptable for the present PoC demonstration but should be addressed in future task-specific validation.

In conclusion, the algorithmic layer is responsible for converting safety information into motion decisions. Obstacle-avoidance planning reduces the probability of entering restricted regions and encountering blocked contact, thereby improving reachability and task continuity. Meanwhile, force-threshold stopping serves as a protective fallback to terminate motion when unexpected contact or blocking occurs. The mechanical and sensing layers provide tool-level baseline protection and online monitoring signals, respectively, which support safe decision-making in the algorithmic layer.

8 | Research Gap and Future Work

Beyond the three-layer safety framework proposed in this work, significant research gaps remain in addressing safety challenges unique to haircutting robots. These gaps highlight the need for systematic studies, experimental validation, and eventual standardization before safe real-world deployment can be achieved.

- **Limitations of the PoC demonstrations.** Current PoC experiments typically rely on static head phantoms and controlled laboratory environments. While such setups are adequate for demonstrating baseline feasibility, they do not capture the inherently dynamic nature of human-robot interaction in real haircutting tasks, where sudden head motions caused by reflexive behaviors, such as coughing or posture adjustment, can introduce rapid and unpredictable disturbances. In addition, experimental validations often simplify safety evaluation at the level of the robot or the end-effector, without explicitly accounting for the geometry and orientation of haircutting tools. Future experimental frameworks should therefore incorporate dynamic head motion, environment-induced disturbances, and safety

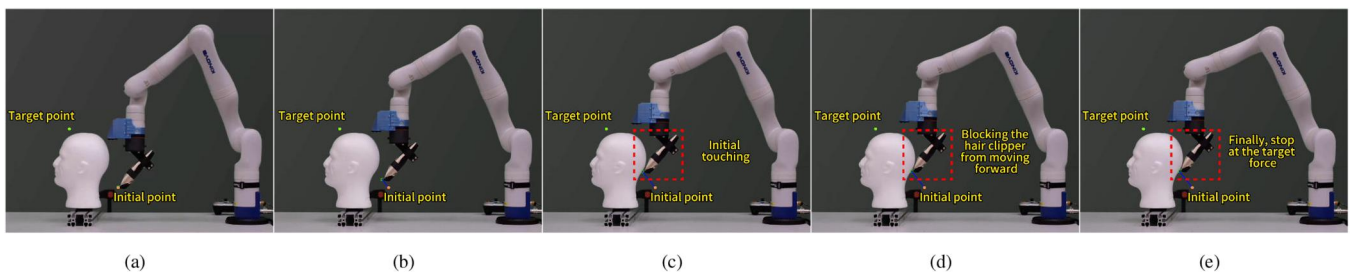


FIGURE 11 | Baseline motion without obstacle-avoidance planning in the haircutting scene. (a)–(e) shows the motion snapshots of the robotic haircutting process. [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1002/rob.70305)]

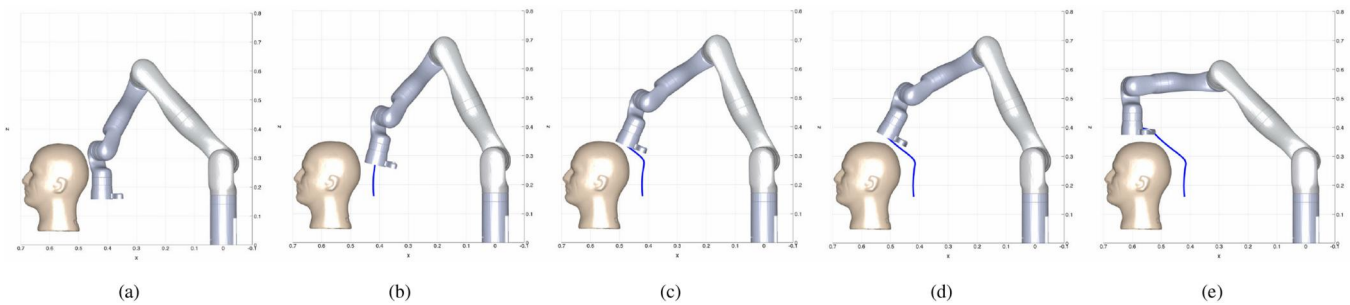


FIGURE 12 | Planned collision-free trajectory in simulation for the Kinova Gen3 manipulator (blue curve). (a) Start configuration. (b)–(d) Intermediate configurations following the planned path while avoiding the head model. (e) Final configuration approaching the target region. [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com)]

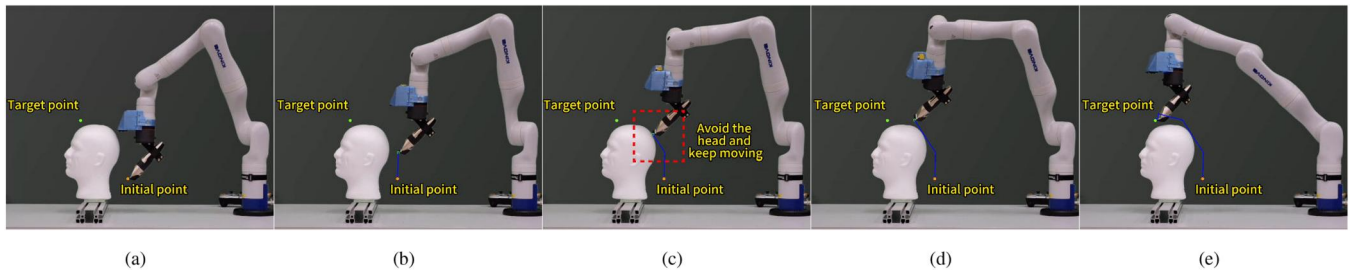


FIGURE 13 | Execution of the obstacle-avoidance motion in the haircutting scene. (a) and (b) The robot starts from the initial point and moves toward the target. (c) The trajectory avoids the head (restricted region) and continues moving. (d) and (e) The robot proceeds along the safe path toward the target region. [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com)]

evaluation that explicitly accounts for tool geometry, in order to better bridge the gap between laboratory demonstrations and real-world robotic haircutting.

- **Lack of quantified safety thresholds.** Unlike industrial collaborative robots, where safety limits on impact forces and permissible contact pressures have been extensively studied, the haircutting scenario lacks corresponding quantitative benchmarks. The scalp is a sensitive organ with heterogeneous tolerance across regions (e.g., temples, nape, crown), yet no systematic measurements exist for defining safe force, torque, or thermal exposure levels during haircutting or styling. Similarly, acceptable latency ranges for remote operation have not been experimentally characterized, leaving uncertainty over how communication delays translate into practical risks. To address this gap, we review relevant international standards and prior studies from adjacent robotic domains, and summarize quantitative reference values from the literature in Table 7. These values are not validated safety limits for robotic haircutting. Instead, they serve as provisional design references for preliminary safety-oriented system development in the absence of guidelines specific to this task. For metrics such as the safety distance between the tool and the skin, as well as the operational distance, no universal numerical limits are available in the literature. These values are inherently task-dependent and should be determined according to the grooming operation, tool geometry, user comfort, and overall safety constraints of the system. Future studies should therefore combine biomechanical experiments, psychophysical assessments, and user studies to establish scientifically grounded safety thresholds. These

values would not only inform controller design but also provide regulators with tangible metrics for safety certification.

- **Insufficient benchmarks and validation tools.** The absence of common benchmarks makes it difficult to compare safety mechanisms across studies. Public datasets capturing hair-scalp boundaries, tool-skin interactions, and multimodal sensor signals during haircutting do not currently exist. Moreover, most experiments rely on ad-hoc test setups, limiting reproducibility. A promising direction is the development of anthropomorphic head phantoms with artificial hair and biofidelic scalp materials. Such platforms could integrate embedded sensors for ground-truth force, temperature, and vibration measurements, enabling both algorithm testing and safety validation under controlled yet realistic conditions. Similarly, simulation environments incorporating realistic hair models would support large-scale training and stress-testing of safety algorithms before physical deployment.
- **Need for personalized safety modeling.** Haircutting tasks inherently involve individual variability. Hair density, curliness, and reflectance affect visual perception, while scalp sensitivity and geometry influence safe force thresholds. A uniform safety model risks either over-restriction, reducing efficiency, or under-protection, compromising safety. Future work should therefore emphasize adaptive and personalized safety modeling. This may involve learning user-specific safety envelopes online, using multimodal sensing (vision, force, thermal), or leveraging pre-interaction calibration routines. Machine learning approaches, including meta-learning or transfer learning,

TABLE 7 | Preliminary safety reference values for robotic haircutting.

Category	Metric	Recommended value	Unit	Description	Source
International standards	Maximum contact force	≤ 5–10	N	Conservative limit to avoid excessive pressure from the clipper blade; ISO/TS 15066 allows up to 130 N for facial regions.	International Organization for Standardization (2016c)
	Maximum contact pressure	≤ 5–10	N/cm ²	Based on clipper head contact area; more conservative than ISO/TS 15066 facial limits (110 – 130 N/cm ²).	International Organization for Standardization (2016c)
	Minimum edge radius	≥ 1	mm	Rounded housing recommended for collaborative tools; ISO 10218 suggests 0.5–2 mm to avoid sharp edges.	International Organization for Standardization (2025)International Organization for Standardization (2011)
Academic research or industrial experience	Safety integrity level	PL d/SIL 2	—	Required performance level for all safety-related robot control functions.	International Organization for Standardization (2023)
	Maximum allowed skin temperature	≤ 42	°C	Human pain threshold is ≈43°C; tissue damage may occur above 44°C.	Martin and Falder (2017)
	Teleoperation latency limit	≤ 200	ms	Human performance degrades above 200–300 ms; lower latency improves predictability and stability.	Sheridan (1992)
	Emergency stop reaction time	≤ 100–150	ms	ISO 10218 defines the principle; the recommended value is based on collaborative robot industrial practice.	International Organization for Standardization (2025). International Organization for Standardization (2011)
	Tool-skin safety distance	Task-specific	mm	Derived from perception error, control error, robot reaction time, and safety margin. ISO 10218 provides the formula; actual values are task-dependent.	International Organization for Standardization (2025). International Organization for Standardization (2011)
	Operational distance	Task-specific	mm	Typical working distance for hair trimming and contour following.	—

Note: The listed values are intended as preliminary design references rather than validated safety limits for robotic haircutting. Their use should be further assessed through experiments specific to this task.

could allow safety parameters to generalize across populations while adapting to individual differences in real time.

- **Towards protocols and standardization.** Although general robotic safety standards such as ISO 10218 and ISO/TS 15066 provide guidelines for industrial and collaborative robots, they do not cover the unique requirements of haircutting robots, which operate directly around the head and face and incorporate sharp or heated tools. Without domain-specific protocols, system validation risks remaining inconsistent. Another relevant point of reference comes from the medical robotics field, where robots also interact with the human body rather than relying on workspace separation for safety. Standards such as IEC 80601-2-77 (International Electrotechnical Commission 2019) illustrate how safety requirements can be formalized when close-contact operation is unavoidable, offering conceptual guidance for haircutting robots, particularly in areas where industrial standards provide limited coverage. However, haircutting robots differ substantially from medical systems in intended use, acceptable risk levels, user expertise, and deployment environments. Consequently, medical-robotics standards cannot be directly applied but can serve as methodological references when deriving requirements tailored for consumer service robots.
- **Ethical and privacy considerations.** Beyond physical safety, haircutting robots must process information that reflects a user's facial structure, scalp condition, and hairstyle preferences. Handling such data raises concerns regarding how it is stored, who has access to it, and what consequences may arise if it is misused. Future systems can mitigate these risks by performing most computations directly on the device, thereby reducing the need to transmit sensitive information externally. This approach can be combined with strategies that limit data collection to what is strictly required for the task and convert any retained information into protected or temporary forms that are removed once a session is completed. Architectures that isolate computation from external networks provide an additional means of reducing exposure of identifiable data. Ethical considerations extend beyond privacy. Decisions must be made about the degree of autonomy a haircutting robot should operate with, the circumstances in which human oversight remains necessary, and how responsibility should be assigned if unexpected events occur. Acceptance of such systems also depends on how individuals and communities perceive robotic involvement in personal appearance. Addressing these issues in future work will require clearer procedures for obtaining user consent, explicit definitions of accountability among system developers and operators, and closer alignment with existing privacy and safety regulations. Integrating these elements into system design will help establish a foundation for responsible and trustworthy deployment of robotic haircutting technologies.

Overall, bridging these gaps will require interdisciplinary efforts across robotics, biomechanics, human factors, and ethics. Addressing them will support the transition of haircutting robots from experimental prototypes to trustworthy service systems, enabling safer and more acceptable deployment in real-world environments.

9 | Conclusion

Robotic haircutting combines the complexity of physical human-robot interaction with task-specific hazards that directly affect user safety and comfort. This paper has examined how risks emerge across mechanical structures, sensing systems, and algorithmic control, and how lessons from related domains can inform their mitigation. Beyond technical safeguards, the path toward real-world deployment requires the establishment of measurable safety thresholds, reproducible testing platforms, and protocols that reflect the unique demands of haircutting tasks. Addressing these issues will enable haircutting robots to evolve from experimental prototypes into practical service systems. Achieving this transition is essential not only for advancing robotics research but also for shaping public acceptance of robots performing close-contact, high-precision tasks in daily life.

Acknowledgments

The authors would like to thank Ville Pitkänen and Toni Tolonen from the University of Oulu, Faculty of Information Technology and Electrical Engineering, for their valuable contributions and insightful discussions related to this work.

Funding

The authors have nothing to report.

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

Data sharing not applicable to this article as no datasets were generated or analyzed during the current study.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section.

Supporting File: rob70305-sup-0001-1HaircuttingSafety.mp4.